Project Design Phase-II: Solution Requirements

Functional & Non-Functional Requirements Specification

Date: 31 January 2026	**Team ID:** NOC-PERF-2026
Project Name: Global Olympic Sports Performance Analysis	**Maximum Marks:** 4 Marks

Functional Requirements (FR)

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority
1	**Data Access & Integration**	Ingest and merge historical Olympic datasets (`athletes` and `medal_results`) spanning 1992–2026 across 60 countries and 50 sports categories, alongside regional sports investment figures.	**High**
2	**Data Processing**	Clean, standardize, and aggregate multi-year records; handle missing values; create normalized ratios for per-athlete yield and investment efficiency.	**High**
3	**Business Analysis**	Calculate key metrics including Avg Medals per Athlete by Sport Category (Rowing: 6.916, Shooting: 6.731), Global Medal Counts (USA: 735, China: 696), Host Medal Dominance (Japan: 105), and Regional Efficiency % (South America: 49.64%).	**High**
4	**Business Insights & Visual Outputs**	Deliver two interactive Tableau dashboards featuring horizontal bar charts, investment vs. medal bubble charts, historical trend lines (1992–2026), and regional efficiency pie/donut charts.	**High**
5	**User Interaction & Exploration**	Enable interactive global filters (Country, Sport Category, Olympic Year, Summer/Winter Games, Region) allowing deep-dive comparative analysis for decision-makers.	**Medium**
6	**Data Update & Reporting**	Support seamless data refresh for newly concluded Olympic Games (e.g., post-2026 update) and export filtered visual reports/stories for executive presentations.	**Medium**

Non-Functional Requirements (NFR)

NFR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Intuitive visualization layout with clear color-coded legends, consistent chart formatting, and responsive filter controls accessible to non-technical stakeholders.
NFR-2	**Security**	Role-based access controls for data editing, compliant data handling for national performance metrics, and secure cloud storage on Tableau Server/Snowflake.
NFR-3	**Reliability**	99.9% data accuracy across aggregated KPIs, ensuring complete consistency between raw dataset rows and Tableau visual summaries.
NFR-4	**Performance**	Dashboard interactive query response time under 2 seconds when toggling country/sport filters across the 34-year historical dataset.
NFR-5	**Availability**	24/7 web access to published Tableau Server/Public dashboards for globally distributed analysts across varying time zones.
NFR-6	**Scalability**	Architecture capable of seamlessly accommodating future Olympic Games data releases beyond 2026 without requiring visual structural redesigns.